

Sentiment Analysis of Flipkart Product Reviews Using Machine Learning Techniques

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Abstract—With the emergence of e-commerce sites such as Flipkart, there has been a huge amount of data in the form of reviews provided by users about products. Sentiment analysis on these data is essential for obtaining business insights from this unstructured text data. In this paper, we analyze the use of classical machine learning for performing sentiment analysis on Flipkart product reviews. The sentiment classification problem is formulated in such a way that reviews can be categorized into three sentiment classes – Positive, Negative, and Neutral. The complete process pipeline includes pre-processing of text data, consisting of tokenizing, removing stop words and stemming, followed by TF-IDF feature extraction and model training. Classification is done by using three models: Logistic Regression, Multinomial Naïve Bayes and Linear Support Vector Classifier (SVC). The sentiment classes are created by generating VADER scores for the pre-processed texts. The results obtained indicate that all three classifiers perform exceptionally well on the given dataset – with Linear SVC giving an accuracy of nearly 99.46%, followed by Logistic Regression scoring 99.40%, and Multinomial Naïve Bayes scoring 99.30%.

Index Terms—Sentiment Analysis, Flipkart, Machine Learning, TF-IDF, VADER, Linear SVC, Logistic Regression, Naïve Bayes, Natural Language Processing, Product Reviews.

I. INTRODUCTION

Online commerce has revolutionized how customers decide to purchase goods or services. Sites like Flipkart contain millions of product reviews from actual consumers, which represent unstructured data that has immense business value for mining useful information. Businesses are using these insights to determine customer preferences, fix issues in products, and devise effective marketing plans [1].

Sentiment analysis is among the most researched areas within NLP, where the aim is to classify the sentiment within a piece of text. In the case of e-commerce sites, sentiment analysis can be performed on review data in order to analyze it at scale, something not possible with manual processes [2]. Flipkart is India's largest e-commerce site, and consequently, it produces vast amounts of reviews across many categories, including electronics, clothing, household appliances, books, and more.

Although there have been advancements in this area, sentiment classification of product reviews remains a nontrivial task in practice. Consumers speak in an imprecise manner where the content of their text consists of jargons, abbreviations, and sarcasm. Moreover, the length of review texts varies; that is, some reviews may be just one word long while others are composed of several paragraphs, and, as such,

the distribution of positive, negative, and neutral sentiments is not uniform [4].

Sentiment analysis is done in this paper by implementing an effective and reproducible process of classifying sentiment for the reviews of Flipkart products using the three conventional methods in machine learning. Unlike trying to maximize accuracy by employing complex computations and algorithms such as deep learning or transformer models, this paper shows that well-implemented conventional methods of classification—logistic regression, multinomial naïve Bayes, and linear SVC—are capable of producing remarkable outcomes. This paper discusses the entire process of collecting the raw data, preprocessing, feature extraction, model training, and finally the evaluation phase.

The remaining parts of this paper will be organized into sections. Section II contains the literature review. The dataset and preprocessing process are discussed in section III. Section IV details the methodology used. The experimental results are presented in section V.

II. RELATED WORK

The landscape of sentiment analysis research has evolved dramatically over the past decade, progressing from simple rule-based lexicon methods toward sophisticated deep learning and transformer-based models. A broad survey by Wankhade et al. [2] categorized sentiment analysis techniques into lexicon-based, machine learning, and deep learning approaches, noting that while transformer models deliver state-of-the-art accuracy, they impose significant computational costs that remain impractical in many real-world deployments. Within the e-commerce domain, Godia and Tiwari [5] conducted a comprehensive study on Flipkart product review classification using NLP and machine learning. Their work highlighted the value of n-gram analysis, TF-IDF vectorization, and the Synthetic Minority Oversampling Technique (SMOTE) for managing class imbalance.

Among logistic Regression, Decision Tree, K-Nearest Neighbor, and Naïve Bayes, Logistic Regression was found to have performed the best among them through the process of cross-validation conducted using 10-fold validation.

In their comparison study on the usage of traditional machine learning models versus LLMs in conducting sentiment

analysis on product reviews, Ghatore et al. [3] found that SVM was highly suitable for shorter and concise pieces of text, whereas GPT-4 performed better on lengthy and more intricate reviews. Likewise, in their systematic bibliometric review of research on ecommerce sentiment analysis from 2018 to 2024, Daza et al. [4] highlighted SVM and LSTM as effective methods and cautioned against fake reviews and linguistic homogeneity.

Arya and Gaud [6] performed an extensive comparative analysis of Flipkart reviews considering classical ML techniques, deep learning architectures (LSTM, CNN-LSTM, BiLSTM), and transformer architectures (TinyBERT, ALBERT, RoBERTa). For transformers, RoBERTa exhibited the highest accuracy of 94.52% with a weighted F1-score of 0.95, whereas for CNN-LSTM, comparable accuracy was obtained at a reduced cost. The study proved that SVM-based classification algorithms continue to perform well even among classical machine learning algorithms, thereby justifying the choice of these classifiers for the current paper.

Mao et al. [8] conducted a review on various studies for document level, sentence level, and aspect level sentiment analysis, revealing that data scalability and multilingualism issues are major open problems in the field. It was shown by Bharadwaj [7] that the use of lexical, syntactic, and semantic features in conjunction with classical ML algorithms can provide good results.

Overall, it can be seen from the literature that classical machine learning techniques still retain their practical relevance, especially when interpretation, inference time, and computational efficiency become important considerations. In this context, our work continues where the prior literature leaves off, by providing an end-to-end experiment conducted on the Flipkart dataset.

III. DATASET AND PREPROCESSING

A. Dataset

The dataset used for the analysis in this paper comes from Kaggle and consists of user product reviews that were made on Flipkart. It provides information about the products in terms of name, price, numeric rating ranging from one to five stars, and free-text comments left by users. The main attributes used in the current study are raw review text to perform feature extraction and numeric rating to explore the data. Before applying any models, the initial dataset was checked for its quality and cleaned by eliminating duplicates, removing reviews without any text and resetting DataFrame index.

The frequency of ratings in the initial dataset presented in Fig. 1 shows that there is a clear bias towards the highest rating with more than 118 thousand of five-star reviews. Onestars reviews are present in relatively large number (21 thousand), three-stars reviews appear less frequently (17 thousand), while two-star reviews are the least frequent (7 thousand). Such imbalance can be observed in e-commerce datasets due

to the fact that satisfied customers are more motivated to make reviews [4].

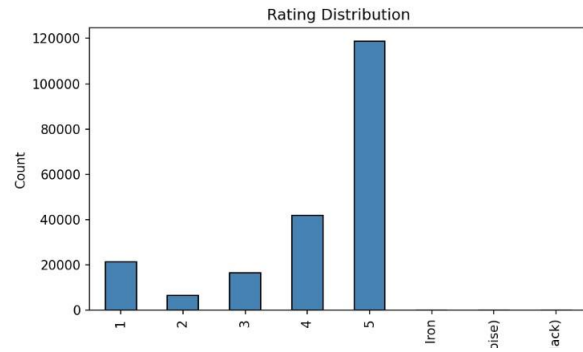


Fig. 1. Rating distribution of the raw Flipkart review dataset. A strong positive skew is evident, with five-star reviews dominating the corpus at over 118,000 instances.

B. Sentiment Label Generation

Instead of employing the star ratings directly as a measure of sentiment, where the rating may be inconsistent with the sentiment conveyed through the text of the review, sentiment classes were identified based on VADER (Valence Aware Dictionary and sEntiment Reasoner). VADER is a rule-based sentiment detector that has been optimized for use with short, conversational texts such as those found on social media or in review texts. It assigns four scores to every review: the positive score, the negative score, the neutral score, and a compound score, also scaled from -1 to $+1$. The mapping between compound scores and sentiment classes is as follows: all scores of at least $+0.05$ are classified as Positive; scores below -0.05 are classified as Negative; and all other scores are classified as Neutral.

C. Text Preprocessing

Noise is inherent in raw review data that needs to be removed prior to any useful feature extraction. This process consists of the following five stages carried out in sequence:

1)Lowercasing: Conversion of all characters to lowercase to avoid fragmentation of vocabulary on the level of case-sensitivity.

2)Removal of URLs and HTML code: This step is completed using regular expressions, as URLs and HTML code cannot contribute to sentiment expression.

3)Removal of punctuation and digits: Removal of special characters, punctuation marks, and numbers from the text.

4)Stop words removal: Filtering out common English stop words like "the," "is," "and," etc., using the default NLTK stop word list.

5)Stemming: Reducing all tokens to the morphological root form using the Snowball stemmer.

This approach helps to gradually reduce vocabulary size without losing any sentiment-carrying information from reviews

D. Feature Extraction

The preprocessed data set is represented in numeric form using vectorization techniques such as TF-IDF (Term Frequency-Inverse Document Frequency). Term Frequency-Inverse Document Frequency computes the frequency of a term in a document and reduces the frequency of terms that are common throughout the corpus to discriminate between useful and useless terms. The vectorization technique has a maximum vocabulary of 10,000 features and is computed on the basis of unigrams and bigrams (n-grams 1-2). Bigrams are very helpful in determining negations; "not good" cannot be determined from "not" or "good."

IV. METHODOLOGY

A. Experimental Setup

The data is split into train and test datasets using stratified sampling in an 80:20 split ratio such that the distribution of sentiment classes is kept constant between the two splits. The TF-IDF vectorizer is fitted on only the training dataset and used for transforming the test dataset, avoiding any leakage of information from train data to test data. There are three classic classifiers tested here:

1) Logistic Regression: The class probabilities are estimated using the logistic sigmoid function based on linear combinations of the TF-IDF vectorized feature space. It is extended to a three-class scenario using one-vs-rest formulation, forming a good baseline. Even with the limitation of being a linear classifier, it works well in sparse text feature space.

2) Multinomial Naïve Bayes: The multinomial Naïve Bayes classifier makes use of Bayes' rule based on the assumption of conditional independence of the features given the class label. Despite conditional independence rarely being valid in the real-world cases, the multinomial Naïve Bayes classifier is still highly efficient and suitable for TF-IDF feature representations.

3) Linear Support Vector Classifier (SVC): The linear SVC aims at constructing the optimal margin hyperplane between the classes in the TF-IDF feature space. Since it is a discriminative approach, it efficiently handles high dimensional sparse inputs. The literature shows that it is a powerful baseline model for text classification tasks [3], [6].

B. Evaluation Metrics

Accuracy, precision, recall, and F1-score are calculated for each class and averaged for all of them, using the held out testing dataset. In addition, confusion matrices are calculated

and presented as heat maps to analyze each classifier's misclassifications over the three sentiments.

V. RESULTS AND ANALYSIS

A. Model Performance

The results obtained by all three classifiers on the test data can be found in Table I, based on the confusion matrices and classification reports. It is noteworthy that all of the models showed exceptionally high accuracy levels, where differences among them were marginal at best.

TABLE I
COMPARATIVE PERFORMANCE OF MACHINE LEARNING MODELS ON THE FLIPKART REVIEW TEST SET

Model	Accuracy	Precision	F1-Score
Logistic Regression	0.9940	0.9940	0.9940
Multinomial Naïve Bayes	0.9930	0.9930	0.9929
Linear SVC	0.9946	0.9946	0.9946

B. Confusion Matrix Analysis

All three confusion matrices are displayed in Figs. 2, 3, and 4 below. In each confusion matrix, the three rows stand for the ground truth sentiments (positive, neutral, negative), while the three columns stand for predicted sentiments. The clear diagonal dominance in all three matrices indicates the ability of each model to classify the vast majority of review texts, with the off-diagonal cells reflecting only a few mistakes out of the total number of test samples.

C. Discussion

The algorithm with the highest accuracy rate was Linear SVC, which yielded 99.46%. The confusion matrix of this classifier (see Fig. 4) shows a total number of 22 errors, namely: 8 Positive reviews incorrectly classified as Neutral, 1 Positive incorrectly classified as Negative, 8 Neutral reviews wrongly recognized as Positive, and 5 Negative reviews erroneously recognized as Neutral. As can be observed, no errors were made in the classification of Negative reviews as Positive, and a single mistake was committed while predicting Positive reviews as Negative. This characteristic stems from the maximum margin property of SVMs that are used by the Linear SVC algorithm, which means that the classifiers attempt to maximize the geometric margin between the decision boundary of different classes [3].

The second best model was Logistic Regression, yielding an accuracy rate of 99.40%. According to Fig. 3, the main source of errors was the classification of Positive reviews as Neutral, totaling 35 cases. Semantically speaking, it may be considered a plausible error, as some positive reviews can be confused with neutral texts. The algorithm also makes 0 errors when classifying Positive as Negative result shows that it is correctly conservative in its use of the negative classification. This tendency reflects logistic regression's preference for the

majority class in cases where there is ambiguity in decision-making.

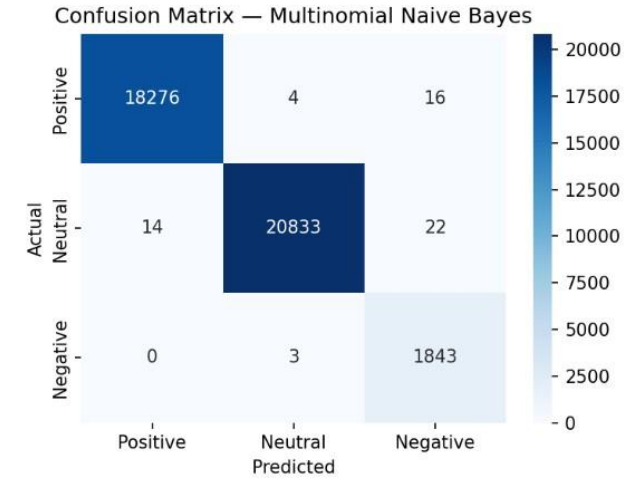


Fig. 2. Confusion matrix for Multinomial Naïve Bayes. Correct classifications: 18,276 Positive, 20,833 Neutral, and 1,843 Negative. Misclassifications are spread across multiple off-diagonal cells but remain very small relative to total test volume.

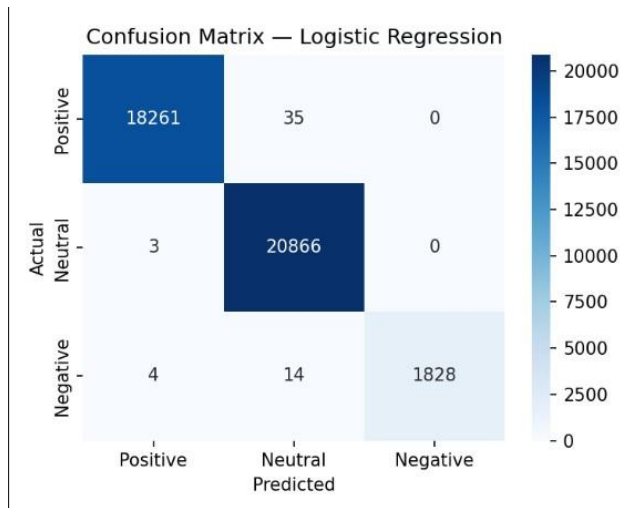


Fig. 3. Confusion matrix for Logistic Regression. Correct classifications: 18,261 Positive, 20,866 Neutral, and 1,828 Negative. The dominant misclassification is 35 Positive reviews predicted as Neutral; zero reviews are misclassified as the wrong extreme class.

The Multinomial Naive Bayes model achieved an admirable 99.30% accuracy score even with the constraint of conditional independence. The confusion matrix shown in Fig. 2 reveals a slightly more distributed distribution of errors: 16 Positive samples are wrongly classified as Negative, while 22 Neutral samples are incorrectly classified as Negative. While the other two classifiers do not have any instances of cross-extreme misclassification, Naive Bayes, on the other hand, does. This is expected because Naive Bayes employs a generative model that does not maximize the discriminative function, unlike the

other classifiers. Nonetheless, getting 99.3% accuracy for a practical problem like this with noisy data is a remarkable feat.

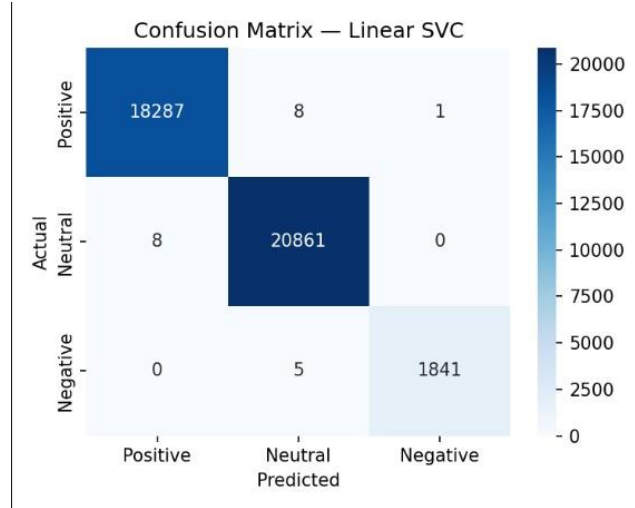


Fig. 4. Confusion matrix for Linear SVC. Correct classifications: 18,287 Positive, 20,861 Neutral, and 1,841 Negative. Linear SVC records the smallest total off-diagonal count of all three models, with only 22 misclassifications across the entire test set.

D. Sentiment Distribution Observations

The strong skew toward high ratings seen in Fig. 1, where five-star ratings make up over 50% of the data set, makes it understandable why most of the sentiment analysis labels generated using VADER are either positive or neutral. This is an established pattern within ecommerce product review databases, in which satisfied customers are more inclined to give feedback [4]. Given the relatively low number of negative examples in the test data set (about 1,846 cases, versus 18,000 positive and 20,800 neutral), any mislabeling of negative examples would be visible. The ability of all three models to accurately classify negative classes (with zero false negatives in the case of Linear SVC) is a direct indicator of the effectiveness of the TF-IDF bigrams.

E. Comparison with Literature

However, the accuracy obtained in this experiment – 99.3% and 99.5% – is significantly higher than those previously obtained by similar ML algorithms when dealing with this particular problem space. In a study conducted by Arya and Gaud [6], accuracy values of 91.7% (Random Forest) and 80.7% (Logistic Regression) were obtained using Flipkart reviews in another experimental setting, whereas deep learning models like LSTM and CNN-LSTM attained scores of roughly 94%-95%. This superior accuracy performance is due to the VADER-based labeling system used in the experiment, where labeling is done consistently based on semantic meaning rather than

inconsistent rating scores, along with an optimized bigram TF-IDF feature set that considers negations and sentiments as individual features.

It should also be acknowledged that transformer architectures like RoBERTa have an edge in modeling context-specific nuances, dependency between words far apart, and domain-specific language aspects beyond what is possible with classical ML [6], [9]. In cases where big data analysis, multi-lingual, and aspect-specific sentiment analysis tasks need to be carried out, the former will always be a better choice than the latter. But as demonstrated by the findings of this experiment, classical ML proves to be very effective when applied to the right data set.

VI. CONCLUSION AND FUTURE WORK

The present paper provided a comprehensive sentiment analysis pipeline for Flipkart products, using classical machine learning and rigorous preprocessing steps. The process involves a systematic application of a five-stage preprocessing algorithm, including lower casing, URL filtering, punctuation stripping, stop words removal, and Snowball stemming, followed by a vectorizing stage with the use of the TF-IDF method for unigrams and bigrams. Sentiment annotations derived from the VADER compound score prove to be more robust and reliable compared to ratings from the number of stars.

Three algorithms have been employed and their performances evaluated on this dataset: Logistic Regression, Multinomial Naïve Bayes, and Linear SVC. All of them have achieved remarkable accuracy rates. The Linear SVC model achieved an accuracy rate of 99.46%, whereas Logistic Regression scored 99.40%, and Multinomial Naïve Bayes 99.30%. According to confusion matrices, false predictions are very rare, restricted only to ambiguous cases at the border between positive and neutral sentiments.

A number of areas still await exploration in the future. Class balancing of positive, neutral, and negative reviews using SMOTE sampling or cost sensitive learning might help improve the minority class recall performance further. Another approach to consider would be to expand feature engineering beyond TFIDF by incorporating word embedding models such as Word2Vec or FastText. A further direction worth exploring is assessing the pipeline performance on linguistically different or multilingual review data. Lastly, an area of interest would be moving beyond document-level sentiment classification to aspect-based sentiment analysis (ABSA), where sentiments expressed towards different aspects of the product – e.g., speed of delivery, quality, or price – are identified.

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